

Lab Report

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Time Spent: 00:09

Score: 0/13 (0%)

Not Passed

Passing Score: 13/13 (100%)

TASK SUMMARY

Required Actions & Questions

On IT-Laptop, launch a DHCP on-path attack using Ettercap

Q1 What is the IP address of the enp2s0 interface on the machine that is initiating the on-path attack?

Your answer:

Correct answer: 192.168.0.46

Capture the DHCP traffic in Wireshark with the bootp filter

On Support: Show Details

Q2 What is the current default gateway for Support?

Your answer:

Correct answer: 192.168.0.5

Q3 What is the default gateway for Support after being caught by the on-path attack?

Your answer:

Correct answer: 192.168.0.46

Q4 Which gateway addresses are provided in the DHCP ACK packets?

Your answer:

Correct answer: 192.168.0.5, 192.168.0.46

On Office1: Show Details

Q5 What is the default gateway for Office1 before renewing its IP addresses?

Your answer:

Correct answer: 192.168.0.5

Q6 What is the first hop in the route to rmksupplies.com?

Your answer:

Correct answer: 192.168.0.5

Q7 What is the default gateway for Office1 after renewing its IP addresses?

Your answer:

Correct answer: 192.168.0.46

Q8 What is the first hop in the route to rmksupplies.com after the on-path attack?

Your answer:

Correct answer: 192.168.0.46

In Office1, log in to the rmksupplies.com Employee Portal

EXPLANATION

Complete this lab as follows:

1. On IT-Laptop, start unified sniffing on the enp2s0 interface.
 - a. From the Favorites bar, select **Ettercap**.
 - b. Select **Sniff > Unified sniffing**.
 - c. From the Network Interface drop-down list, select **enp2s0**.
 - d. Select **OK**.
 - e. Select **Mitm > DHCP spoofing**.
 - f. In the *Netmask* field, enter **255.255.255.0**
 - g. In the *DNS Server IP* field, enter **192.168.0.11**
 - h. Select **OK**.
2. View the IP address of the IT-Laptop computer.
 - a. From the Favorites bar, select **Terminal**.

- b. At the prompt, type **ip addr show** and press **Enter**.
 - c. In the top right, select **Answer Questions**.
 - d. Answer Question 1.
 - e. Close the Lab Questions dialog.
3. On Support, start a Wireshark capture on the enp2s0 interface and filter for only DHCP packets.
 - a. From the top navigation tabs, select **Floor 1 Overview**.
 - b. Under Support Office, select **Support**.
 - c. From the Favorites bar, select **Wireshark**.
 - d. Under Capture, select **enp2s0**.
 - e. Select the **blue fin** to start a Wireshark capture.
 - f. In the *Apply a display filter* field, type **bootp** and press **Enter** to show only DHCP packets.
Notice that no packets are being captured.
4. View the default gateway for Support.
 - a. From the Favorites bar, select **Terminal**.
 - b. Type **route** and press **Enter**.
 - c. In the top right, select **Answer Questions**.
 - d. Answer Question 2.
5. Request new DHCP addresses for the enp2s0 interface.
 - a. From the terminal, type **ip link set enp2s0 down** and press **Enter** to bring the interface down.
 - b. Type **ip link set enp2s0 up** and press **Enter** to bring the interface back up.
Notice that DHCP packets (bootp) have been captured.
6. View the current default gateway on Support.
 - a. In Terminal at the prompt, type **route** and press **Enter**.
 - b. Answer Question 3.
The on-path attack has captured your session and is now acting as an intermediary router.
 - c. Close the Lab Questions dialog.
7. Analyze the DHCP ACK packets to find the router IP addresses used.
 - a. Maximize Wireshark for easier viewing.

In Wireshark, under the Info column, notice that there are two DHCP ACK packets. One is the real acknowledgment (ACK) packet from the DHCP server, and the other is the spoofed ACK packet.

- b. Select the first **DHCP ACK** packet received.
 - c. From the middle panel, expand **Bootstrap Protocol (ACK)**.
 - d. Expand **Option: (3) Router**.
Make note of the router IP address.
 - e. Select the second **DHCP ACK** packet received.
 - f. Under *Option: (3) Router*, notice the IP address for the router.
 - g. In the top right, select **Answer Questions**.
 - h. Answer Question 4.
 - i. Close the Lab Questions dialog.
8. In Office1, view the current default gateway and the route to the rmksupplies.com site.
 - a. From top navigation tabs, select **Floor 1 Overview**.
 - b. Under Office 1, select **Office1**.
 - c. Right-click **Start** and select **Windows PowerShell (Admin)**.
 - d. Type **ipconfig** and press **Enter**.
 - e. In the top right, select **Answer Questions**.
 - f. Answer Question 5.
 - g. Type **tracert rmksupplies.com** and press **Enter**.
 - h. Answer Question 6.
Note that the first hop and default gateway were determined prior to the on-path attack.
9. View the effects of the on-path attack on Office1's IP addresses.
 - a. At the prompt type, **ipconfig /release** and press **Enter** to release the currently assigned addresses.
 - b. Type **cls** and press **Enter** to clear the screen.
 - c. Type **ipconfig /renew** and press **Enter** to request new IP addresses from the DHCP server.
 - d. Answer Question 7.
Notice that the default gateway has changed to the attacker's computer, which has an IP address.

- e. Type **tracert rmksupplies.com** and press **Enter**.
 - f. Answer Question 8.
 - Notice that the first hop is now using the attacker's IP address and then goes to the correct default gateway of 192.168.0.5.
 - g. Close the Lab Questions dialog.
10. From Google Chrome, log into the rmksupplies.com Employee Portal.
- a. From the taskbar, select **Google Chrome**.
 - b. Maximize the window for easier viewing.
 - c. In the URL field, enter **rmksupplies.com** and press **Enter**.
 - d. At the bottom of the page, select **Employee Portal**.
 - e. In the Username field, enter **bpascal**.
 - f. In the Password field, enter **Dontrygu3ss1ng**.
 - g. Select **Login**. (You are logged in as Blaise Pascal.)
11. From IT-Laptop, find the captured username and password in Ettercap.
- a. From the top navigation tabs, select **Floor 1 Overview**.
 - b. Under IT Administration, select **IT-Laptop**.
 - c. Maximize Ettercap.
 - d. In Ettercap's bottom pane, find the username and password used to log in to the Employee Portal.
12. Answer the questions.
- a. In the top right, select **Answer Questions** to end the lab.
 - b. Select **Score Lab**.